

Li Bai the Youth: An LLM-Powered Virtual Agent for Children's Chinese Poetry Education

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Abstract

Large language models (LLMs) are increasingly recognized for their potential, but their application in fostering children's poetry learning remains underexplored. This paper introduces *Li Bai the Youth*, an interactive installation that leverages a virtual agent powered by an LLM. We present a technical pipeline incorporating text-to-speech, lip synchronization, and other functionalities to enable real-time conversation between users and the virtual agent, Li Bai. Additionally, we implemented strategies to enhance the user experience during interaction. Li Bai the Youth offers a novel approach to poetry learning, promoting cultural heritage engagement.

Keywords

Large Language Models, Virtual Agent, Poetry Education, Real-time Conversation

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1 Introduction

Virtual agents are increasingly popular for enhancing student engagement and interactivity in education. Baylor highlights their effectiveness in promoting motivation and positive learning attitudes [Baylor 2009]. Hew et al. explore their functions, such as providing feedback, answering questions, and delivering content engagingly [Hew et al. 2023]. These studies suggest virtual agents play a key role in creating immersive, interactive learning environments. LLMs have also become powerful educational tools, processing and generating text for personalized learning. Meyer et al. show

LLMs effectively tailor materials and provide individualized feedback [Meyer et al. 2024]. Kazemitabaar et al. explore their potential in automated tutoring, offering students on-demand support and guidance [Kazemitabaar et al. 2024].

2 Related Work

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3 Design and Implementation

3.1 System Design

In designing the character of *Li Bai the Youth*, we used AI to generate numerous facial and clothing designs. After selecting and combining these elements, we created a 2D concept image. For the 3D model, we referenced popular historical dramas to align with modern aesthetic preferences. To ensure that Li Bai's appearance reflects Tang Dynasty characteristics, we conducted in-depth research on Tang attire and made detailed adjustments, such as retaining the round collar design. This approach blends historical authenticity with modern appeal, attracting young audiences to interact with him and sparking their interest in classical poetry.

The installation of *Li Bai the Youth* features a high-performance PC, OLED TV, and microphone, enabling interaction for individuals and groups. Participants take turns engaging in poetic dialogue with Li Bai, while others observe and enjoy the experience.

3.2 Technical Pipeline

Speech Recognition. User speech input via the microphone is converted into text by the system's ASR module for further processing. We integrated Tencent Cloud ASR via an Unreal Engine

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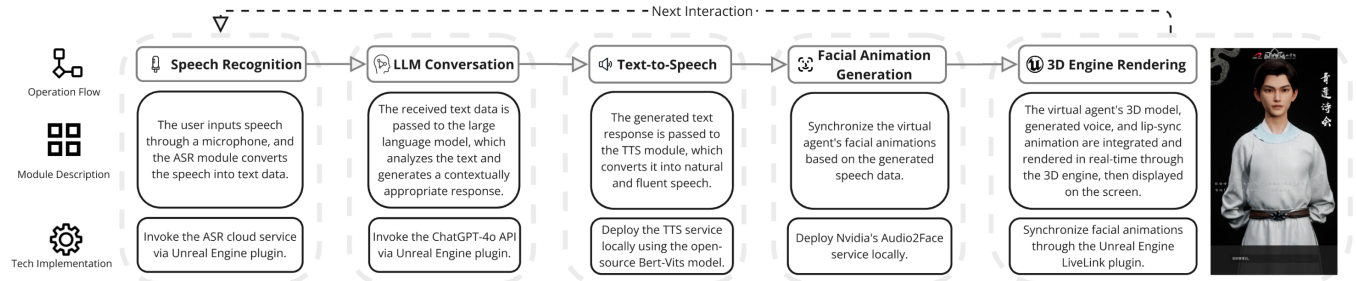


Figure 1: The technical pipeline of the system.

plugin to ensure efficient, accurate speech recognition, optimized for real-time performance.

Large Language Model Response. User input is processed by the large language model, which generates contextually appropriate responses. We use the ChatGPT-4 API via an Unreal Engine plugin, configuring the system to define Li Bai’s character and poetry dialogue functions, ensuring response quality and coherence.

Text-to-Speech (TTS). We deployed a locally hosted open-source Bert-VITS model for TTS and used Unreal Engine to send HTTP requests to the local API. Bert-VITS combines BERT’s text comprehension with VITS’s speech synthesis, producing natural, fluent, and emotionally rich speech for the virtual agent’s voice.

Facial Animation Generation. Facial animations are generated by a locally deployed Nvidia Omniverse Audio2Face server. After speech is produced, the Python server calls the Audio2Face API to synchronize the animations and automatically adjust emotions based on the speech, ensuring the virtual agent’s facial movements align with the dialogue.

3D Engine Rendering. We integrate the virtual agent’s 3D model, voice, and facial animations, rendering them in real-time through Unreal Engine and displaying them on screen. This process uses the Unreal Engine LiveLink plugin to transmit animation data from Audio2Face in real-time, ensuring synchronized rendering.

3.3 Optimization

Large Language Model Response Optimization. To enhance the quality of the large language model’s responses, we adopted the CO-STAR framework, developed by GovTech’s Data Science and AI team. This framework helped us optimize prompt design by considering key factors affecting LLM response quality. It allowed us to not only adjust the language style but also limit the word count, ensuring smooth dialogue flow and improving the user experience.

TTS Process Efficiency Improvement. To reduce text-to-speech conversion time, we implemented text segmentation and parallel processing. The system first splits the response into sentences, prioritizing the first for speech conversion and facial animation, while the remaining sentences are processed subsequently. This approach provides faster feedback, even as the rest of the speech is still being generated.

Dynamic Visual Feedback Optimization. To reduce perceived delay, we designed dynamic visual feedback. When users speak, Li Bai performs a listening gesture; as the large language model responds, text gradually appears, and the virtual camera shifts from a full-body to a half-body view. These changes minimize perceived system delay and enhance overall smoothness and immersion.

4 Evaluation

To enhance students’ engagement with classical poetry, we partnered with the Affiliated Middle School of the Communication University of China to introduce the Li Bai the Youth system into classrooms. Students were asked to interact with the virtual agent, engaging in poetic dialogue with Li Bai. Initial feedback from five students indicated that they found the experience enjoyable and engaging. In the future, we plan to conduct formal qualitative and quantitative evaluations to assess the system’s impact.

5 Conclusion

Our research integrated a virtual character powered by a large language model, transforming one-way teaching into an interactive, multimodal experience. Initial results show the system enhances student engagement and interest in poetry. This work lays the foundation for future studies using AI-powered virtual agents to enrich education.

Acknowledgments

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